# The paleomagnetism of the Croker Island Complex, Ontario, Canada

H. C. PALMER
*Department of Geophysics, The University of Western Ontario, London, Ontario*
Received October 24, 1968
Accepted for publication January 10, 1969

Meaningful paleomagnetic results were obtained from 23 of 30 oriented samples from the Croker Island Complex, North Channel of Lake Huron, Canada. The age of the complex is estimated to be 1475 ± 50 m.y. from a Rb–Sr isochron. The paleomagnetic pole calculated from the directions of remanent magnetization after partial ac demagnetization is 143° West, 5½° North. This result, when compared with other paleomagnetic results from North America, indicates little or no polar wandering in the time interval 1.48 b.y. to 1.1 b.y. However, a significant shift in the pole position is indicated during the time interval 1.7 to 1.48 b.y.

## Introduction

A considerable number of paleomagnetic studies of North American Precambrian rock units have been reported in the literature. However, the geological age of most of the units studied is presently known only within wide limits. The construction of a North American Precambrian polar wandering curve will require many more paleomagnetic results from rock units that have been radiometrically dated. In the case of igneous rocks, units should be chosen for which the radiometric 'dates' are thought to be approximations to the geological 'age', i.e., rocks for which the post-crystallization thermal history has been simple. The Croker Island Complex, exposed on several islands in the North Channel of Lake Huron, appears to satisfy the above requirement.

## Geology and Age

The Croker Island Complex is circular in plan with a diameter of about 6½ miles as estimated from the aeromagnetic contours (Fig. 1). The geology of the complex has been described by Card (1965) and the following is extracted from his paper. "The outermost rocks of the intrusive body are granitic and syenitic, with narrow gabbro layers. The core of the complex consists of rocks of intermediate and mafic compositions." The weight of evidence indicates that the pluton was emplaced after the major periods of folding that affected the Huronian (?) rocks into which it was intruded. There is no evidence to suggest that the complex underwent any postemplacement deformation or reheating. The estimated trace of the Grenville Front lies some 30 miles to the southeast, as deduced from a study of the magnetic provinces of Lake Huron (Nwachukwu *et al.* 1965).

Van Schmus (1965) reported a Rb–Sr isochron date of 1475 ± 50 m.y. (from granitic rocks of the complex), which he interpreted as a valid estimate of its age.

## Sampling and Remanent Magnetization Measurements

Thirty oriented block samples comprising 10 gabbroic and dioritic rocks and 20 syenodioritic to granitic rocks were collected from 19 sites (Fig. 1) within the complex. Azimuthal orientations were made with a Brunton compass checked against the bearings to landmarks identified on a topographic map of the area.

In the laboratory a 3.5 cm diameter core was drilled from each sample and the cores were cut in lengths of 3.5 cm to provide two or more specimens for measurement on an astatic magnetometer. The natural remnant magnetization (NRM) directions of 28 of the 30 samples could be measured on the astatic system. The specimens of these 28 samples were demagnetized in an ac field of 50 Oe rms (root mean square). The directions of magnetization of most of the specimens could still be measured after this treatment and the results were averaged vectorially to give sample mean directions. In the few cases where the intensity of magnetization after this first stage of demagnetization was below the limit of measurement of the magnetometer, the undemagnetized direction of that sample was used in the calculation of the population mean. A summary of these results appears in the third row of Table I.

Inasmuch as there was no confidence that a

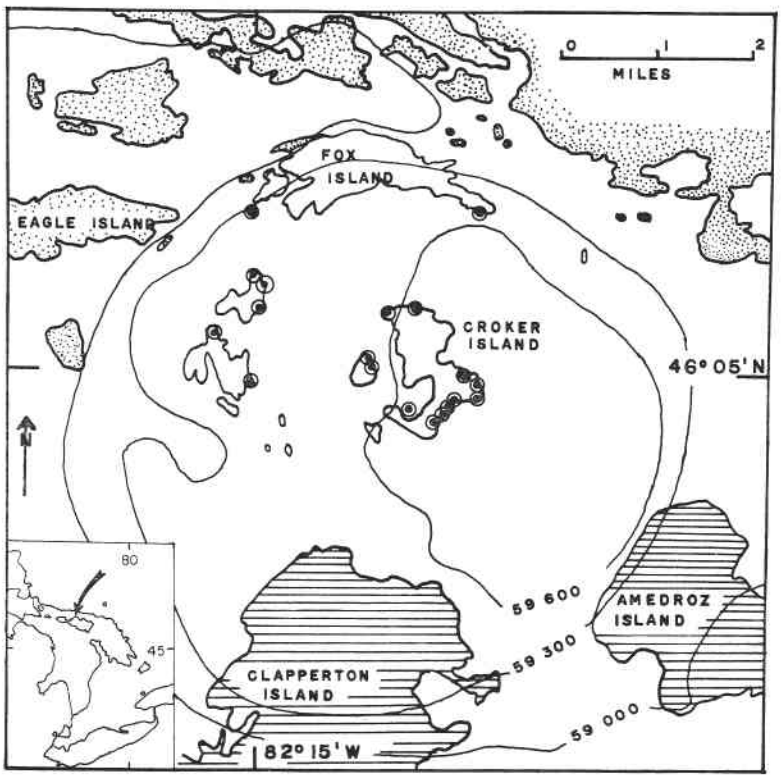



FIG. 1. Generalized geological map, showing sampling sites (large dots) of the Croker Island Complex, Lake Huron, Ontario. Older rocks are stippled and younger Paleozoic rocks are ruled. Total magnetic contours, with values in gammas, outline the circular plan of the complex. Map prepared from Card (1965). Inset map shows general location of the Croker Island Complex in Lake Huron.

TABLE I

Summary of paleomagnetic data

| Measuring instrument | Bac | D° | I° | N | E | *R* | *k* | α95 | Pole position Long (°W) | Lat.(°N) | dp | dm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Spinner | NRM | 269.8 | +54.2 | 29 | 1 | 24.60 | 6.4 | | | | | |
| Spinner | 50 | 258.3 | +52.3 | 29 | 1 | 24.49 | 6.2 | | | | | |
| Astatic | 50 | 257.8 | +54.2 | 28 | 2 | 25.81 | 12.3 | | | | | |
| Spinner | 100 | 261.4 | +49.5 | 29 | 1 | 24.27 | 5.9 | | | | | |
| Spinner | 150 | 256.6 | +48.7 | 29 | 1 | 25.23 | 7.4 | | | | | |
| Spinner | 200 | 256.2 | +48.0 | 29 | 1 | 26.05 | 9.5 | | | | | |
| Spinner | 300 | 252.3 | +50.2 | 29 | 1 | 25.15 | 7.3 | | | | | |
| Spinner | 200–300 | 256.4 | +48.4 | 29 | 1 | 26.31 | 10.4 | 8.6° | 142.3° | 12.2° | 7.7 | 11.6 |
| Spinner | 200–300 | 251.7 | +41.6 | 23 | 7 | 21.40 | 13.7 | 8.5° | 142.8° | 5.4° | 6.7 | 10.5 |

NOTE: Bac = strength of alternating field in oersteds rms.
D = declination in degrees east of north.
I = inclination in degrees (positive downward).
N = number of samples included in the statistical analysis.
E = number of samples excluded in the statistical analysis.
R = resultant of the N unit vectors (Fisher 1953).
*k* = best estimate of precision (Fisher 1953).
α95 = semi-angle of cone of 95% confidence for mean direction.
dp = is the semi-axis of the oval of 95% confidence for the pole position, along the great circle through the site.
dm = is the other semi-axis.

stable end point for the mean direction had been reached after this first stage of ac demagnetization, additional cores were drilled from the samples to provide 2 to 4 specimens of 2.5 cm diameter per sample. The NRM directions and intensities of these specimens were measured on a more sensitive spinner magnetometer (Princeton Applied Research Model SM-1). The results of these measurements were averaged vectorially to give sample mean directions, which are plotted in Fig. 2, and the sample means were averaged

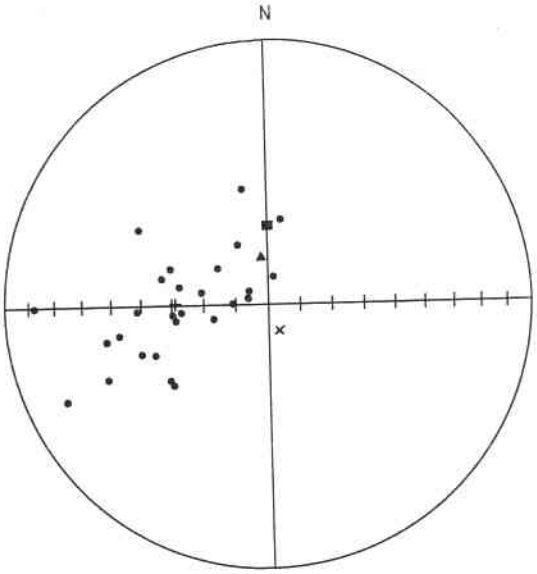



FIG. 2. NRM sample directions of Croker Island Complex. Conventions of the equal-area projection: N = geographic north; ● = north-seeking directions downward; × = north-seeking direction upward; + = mean direction of magnetization; ■ = axial dipole field direction; ▲ = present field direction.

to give the results listed in Table I. Specimens were variable in initial intensity of magnetization, ranging from $9.0 \times 10^{-3}$ to $1.0 \times 10^{-6}$ emu $cm^{-3}$. In one case, where there was no agreement among the directions of the individual specimens, the sample was omitted from the calculation of the mean direction.

### Alternating Field 'Cleaning'

Each of the spinner-size specimens was subjected to ac demagnetization at 50 Oe rms, 100 Oe rms, 150 Oe rms, 200 Oe rms, and 300 Oe rms and remeasured after each stage of demagnetization. Sample mean directions were calcu-

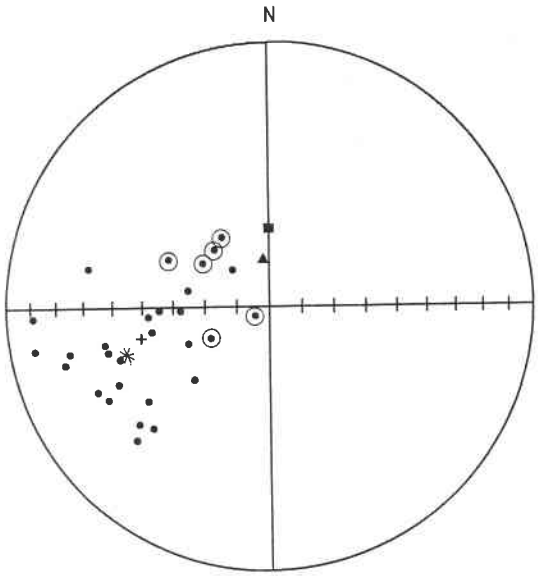



FIG. 3. Sample directions after 200–300 Oe r.m.s. treatment. + = mean direction of all samples; * = mean direction when circled samples are excluded from the analysis. Other conventions are as in Fig. 2.

lated as before and used to calculate the results listed in Table I. Maximum precision of the mean direction could be obtained by using the results after either the 200 or 300 Oe field treatment; the choice for a particular sample was based on whichever treatment gave the maximum within-sample precision estimate "$k$" (Fisher 1953). These results are plotted in Fig. 3, and the mean direction with its associated statistics is recorded in the second last row of Table I.

The second and third rows of Table I compare the results obtained on the spinner magnetometer with those obtained using the astatic magnetometer. The astatic magnetometer has been in use in our laboratory for a number of years, whereas this study incorporates the first published results using the spinner magnetometer. The results are not meant to be a rigorous inter-instrument test inasmuch as the astatic results include those of a few undemagnetized samples, the sample mean directions were not based on the same specimens, and the total number of samples is not equal. The results do, however, suggest that there is no large systematic difference in directions obtained in using either instrument. The astatic results are not considered further in this investigation.

An inspection of Figs. 2 and 3 shows that the precision of the mean direction is increased after

the ac demagnetization, but that the dispersion is still relatively high and that the directions tend to have a planar distribution towards the present field direction rather than a Fisherian distribution about the mean direction. Plots made of the intensity of magnetization after each stage of demagnetization, calculated as a percentage of the original intensity, were made for all the samples. The variation in coercive force spectrum was very large. One sample displayed essentially no decrease in intensity with demagnetization, others moderate amounts of decrease, while still others decreased to about 2% of their original intensity. There appears to be no correlation between type of demagnetization curve and proximity of direction to that of the present field at the collecting site. There is a correlation, however, of within-sample scatter of directions and proximity to the present field direction. Each sample mean direction plotted in Fig. 3 is defined with highly variable precision; the within-sample precision estimate $k$ (Fisher 1953) ranges from 2.8 to 693. Samples having within-sample precision estimates less than 10 (the precision estimate $k$ for the population of 29 samples as a whole) have been circled in Fig. 3. All six samples have steep positive inclinations, which suggests that even after partial demagnetization these inhomogeneously magnetized samples still retain a large present field component of magnetization. The precision is increased slightly by excluding these samples from the statistical analysis and their exclusion probably results in a more accurate estimate of the paleomagnetic field direction. This direction and its corresponding pole position are recorded in the last row of Table I.

## Pole Positions

In Fig. 4 the Croker Island pole position is compared with other pole positions previously reported in the literature. In some cases the estimates of the geological age have been reported subsequent to the publication of the paleomagnetic results, and it seems worthwhile to summarize the results from North American rock units that are close in age to the Croker Island Complex.

The Croker Island pole position is very close to the pole positions (1 and 2 of Fig. 4) calculated by Hays and Scharon (1966) and Hsu *et al.* (1966) from Precambrian rocks of Missouri. Mineral ages from granites and pegmatites, some of which

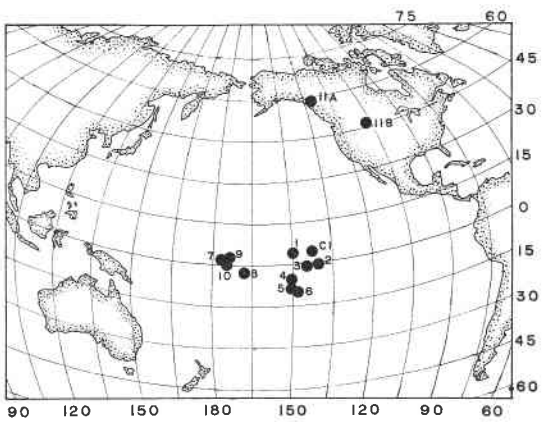



FIG. 4. Paleomagnetic poles derived from North American rocks. The Croker Island Complex pole (CI) and poles from 1 to 10 fall in the age range 1475 m.y. to 1100 m.y. Pole 11 is from the Sudbury irruptive of 1700 m.y.

intrude the rocks investigated paleomagnetically, range from 1300 m.y. to 1425 m.y. (Tilton *et al.* 1962), which suggests a minimum age for the Missouri pole positions. Recently Bickford and Odom (1968) reported a 1315 ± 35 m.y. Rb–Sr isochron from the Stout's Creek Rhyolite and a 1420 ± 30 m.y. Rb–Sr isochron from the Hogan Mountain and Royal Gorge Rhyolites. These rock units constitute some of the units studied paleomagnetically by Hays and Scharon (1966). Thus, if the age of magnetization can be correlated with the geological age of these rock units, then the Missouri paleomagnetic result has sampled the field direction at least twice in a 100 m.y. time interval.

Pole 3 is that calculated from the Michikamau anorthositic intrusion, Labrador, recently reported by Murthy *et al.* (1968). These authors cited the evidence for assigning a 1400 ± 80 m.y. age for this pole position.

Pole 4 is that calculated from normal and reversed directions of magnetization from several rock types of 1410 ± 30 m.y. age from the Front Range of Colorado–Wyoming (Eggler and Larsen 1968).

Pole 5 is the 'Belt Series of Montana' combined pole taken from Irving (1964), which was computed from the results of Collinson and Runcorn (1960). The formations represented by the mean Belt pole range from the Appekunny (oldest) to the McNamara (youngest). According to the correlations given by Smith and Barnes (1966) and the Rb–Sr isochron work of Obradovich and

Peterman (1968), the Belt pole combines results spanning the time interval 1325 ± 15 m.y. to 1100 ± 53 m.y. These results are compatible with the 1085 ± 52 m.y. K–Ar dates reported by Hunt (1962), which set a minimum age for the Purcell pole position of Black (1963), which is plotted as pole 6 of Fig. 4.

Pole 7 is that calculated from directions of magnetization of the MacKenzie dike swarm, and pole 8 is that obtained from the Sudbury dike swarm. Fahrig *et al.* (1965), in presenting preliminary paleomagnetic data for these rocks, reported K–Ar results that suggest ages of 1295 m.y. and 1285 m.y. for the MacKenzie and Sudbury swarms respectively. The Sudbury dike pole plotted in Fig. 4 is that given by Larochelle (1967).

Poles 9 and 10 are those reported by Robertson (1964) for the Muskox gabbroic rocks and the Coppermine lavas respectively. Robertson quoted ages of 1150 ± 40 and 1150 ± 100 m.y. for these rock units.

Poles 11A and 11B are those reported by Sopher (1963) for the Sudbury irruptive. Both poles are calculated from the same data; pole 11B assumes a 10° rotation of the north range, whereas pole 11A does not. Fairbairn *et al.* (1968) reported a 1704 ± 19 m.y. Rb–Sr isochron for the norite–granophyre intrusive pair of the irruptive.

## Discussion

The age of the Croker Island Complex and the ages of the rock units yielding poles 1 to 4 of Fig. 4 are very similar. The agreement of the pole positions calculated from these practically concordant units from widely separated regions of North America support the axially centered dipole model assumed in paleomagnetic work. The close grouping of these pole positions further suggests that there has been no relative rotation of crustal blocks from these widely separated sampling regions of North America since this epoch in Precambrian time.

A further point of interest revealed by Fig. 4 is that from 1475 m.y. to about 1100 m.y. there appears to have been little polar movement with respect to North America, but that there was a large shift in the position of the poles in the shorter time interval 1700 m.y. to 1475 m.y. An obvious omission from Fig. 4 are the Middle Keweenawan poles reported by DuBois (1962). A study I have completed of the Middle Keweenawan volcanic sequences of the Lake Superior region essentially confirms many of DuBois' pole positions in the north Pacific. The problems posed by relating the Middle Keweenawan poles to those of the Belt–Purcell–Coppermine lavas cannot be adequately evaluated until radiometric age dating on the Keweenawan volcanics, which is in progress in several laboratories, is completed.

## Acknowledgments

K. D. Card suggested the possibility of studying the paleomagnetism of the complex. C. M. Carmichael and G. M. Young offered constructive criticism of the manuscript. John Starkey kindly supplied a plotter program, and Lynda Parker modified this program, which produced the equal-area nets used in this paper.

The field work was completed while the author was a recipient of a special fellowship in science from the Canada Council. The laboratory work and data processing costs were met by a grant from the National Research Council of Canada.

Bickford, M. E. and Odom, A. L. 1968. Rb–Sr geochronology of igneous events in the Precambrian of the St. Francis Mountains, southeastern Missouri (Abstr.). Program with Abstracts, 1968 Ann. Meet. Geol. Soc. Amer., Mexico City, p. 27.

Black, R. F. 1963. Paleomagnetism of part of the Purcell System in southwestern Alberta and southeastern British Columbia. Geol. Surv. Can., Bull. 34.

Card, K. D. 1965. The Croker Island complex. Ont. Dept. Mines Geol. Circ. No. 14.

Collinson, D. W. and Runcorn, S. K. 1960. Paleomagnetic observations in the United States: new evidence for polar wandering and continental drift. Bull. Geol. Soc. Amer. **71**, pp. 915–958.

DuBois, P. M. 1962. Paleomagnetism and correlation of Keweenawan rocks. Geol. Surv. Can., Bull. 71.

Eggler, D. H. and Larsen, E. E. 1968. Paleomagnetic study of dated Precambrian rocks of the Front Range, Colorado–Wyoming. Geophys. J. Roy. Astron. Soc., **14**, pp. 497–504.

Fahrig, W. F., Gaucher, E. H., and Larochelle, A. 1965. Paleomagnetism of diabase dykes of the Canadian Shield. Can. J. Earth Sci., **2**, pp. 278–298.

Fairbairn, H. W., Faure, G., Pinson, W. H., Hurley, P. M. 1968. Rb–Sr whole-rock age of the Sudbury lopolith and basin sediments. Can. J. Earth Sci., **5**, pp. 707–714.

Fisher, R. 1953. Dispersion of a sphere. Proc. Roy. Soc. London, A, **217**, pp. 295–305.

Hays, W. W. and Scharon, L. 1966. A paleomagnetic investigation of some of the Precambrian igneous rocks of southeast Missouri. J. Geophys. Res., **71**, pp. 553–560.

Hsu, I.-Chi, Anderson, R. E., and Scharon, L. 1966. Paleomagnetic properties of some of the Precambrian rocks in Missouri. J. Geophys. Res., **71**, pp. 2645–2650.

Hunt, Graham. 1962. Time of Purcell eruption in southeastern British Columbia and southwestern Alberta. J. Alberta Soc. Petrol. Geol., **10**, pp. 438–442.

IRVING, E. 1964. Paleomagnetism and its application to geological and geophysical problems. John Wiley and Sons, New York.

LAROCHELLE, A. 1967. The paleomagnetism of the Sudbury diabase dike swarm. Can. J. Earth Sci., **4**, pp. 323–332.

MURTHY, G. S., FAHRIG, W. F., and JONES, D. L. 1968. The paleomagnetism of the Michikamau anorthositic intrusion, Labrador. Can. J. Earth Sci., **5**, pp. 1139–1144.

NWACHUKWU, S. O., BECK, A. E., and CURRIE, J. B. 1965. Magnetic provinces of Lake Huron and adjacent areas and their geologic significance. Can. J. Earth Sci., **2**, pp. 227–236.

OBRADOVICH, J. D. and PETERMAN, Z. E. 1968. Geochronology of the Belt Series, Montana. Can. J. Earth Sci., **5**, pp. 737–747.

ROBERTSON, W. A. 1964. Paleomagnetic results from northern Canada suggesting a tropical Proterozoic climate. Nature, **204**, No. 4953, pp. 66–67.

SMITH, A. G. and BARNES, W. C. 1966. Correlation of and facies changes in the carbonaceous, calcareous, and dolomitic formations of the Precambrian Belt–Purcell Supergroup. Bull. Geol. Soc. Amer., **77**, pp. 1399–1426.

SOPHER, S. R. 1963. Paleomagnetic study of the Sudbury irruptive. Geol. Surv. Can., Bull. 90.

TILTON, G. R., WETHERILL, G. W., and DAVIS, G. L. 1962. Mineral ages from the Wichita and Arbuckle Mountains, Oklahoma, and the St. Francis Mountains, Missouri. J. Geophys. Res., **67**, pp. 4011–4019.

VAN SCHMUS, R. 1965. The geochronology of the Blind River – Bruce Mines area, Ontario, Canada. J. Geol., **73**, pp. 755–780.